

Two-Stage Semantic-Aware Visual Prompting for Object Grounding in Robotics

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Abstract—We propose a lightweight two-stage framework for visual grounding in robotics, combining semantic-aware candidate extraction with visual prompting. The first stage identifies object candidates using efficient segmentation and detection models. In the second stage, region cropping and multi-level Gaussian blur enhance visual features for better alignment with language queries via CLIP. Without any fine-tuning, our method achieves competitive accuracy on RefCOCO, RefCOCO+, and RefCOCOg. The framework is well-suited for real-time, resource-constrained robotic systems.

Index Terms—Visual Grounding, Visual Prompting, CLIP, Object Localization, Robotics

I. INTRODUCTION

Visual grounding plays a critical role in enabling robots to perceive and interact with their environment through natural language. It involves identifying the location of a target object in an image, given a referring expression such as “the red mug on the table.” This task is especially important in robotics, where accurate object localization from a first-person perspective is essential for manipulation, navigation, and human-robot interaction.

Although recent progress in vision-language models (VLMs), particularly CLIP (Contrastive Language–Image Pre-training), has significantly advanced cross-modal understanding, existing visual grounding methods still face several practical limitations. Many approaches rely heavily on end-to-end supervised training with large-scale annotated datasets, which restricts their generalization ability to unseen objects or domains. Furthermore, in complex real-world environments, these models often struggle with cluttered scenes, ambiguous object boundaries, and background distractions. As a result, their localization accuracy degrades when visual and semantic signals become noisy or less distinct. In addition, the computational overhead of some state-of-the-art models makes them unsuitable for real-time deployment in resource-constrained robotic platforms.

To address these limitations, we propose a two-stage semantic-aware visual prompting framework designed to improve grounding performance in real-world robotics applications. In the first stage, semantic segmentation and object detection are used to extract candidate object regions from an RGB image, effectively narrowing the search space and reducing background interference. In the second stage, we apply visual prompting techniques—such as region cropping and multi-level Gaussian blur—to enhance the semantic saliency of each candidate region. These processed visual inputs are then fed into the CLIP model for cross-modal alignment with

the textual description. The cosine similarity between image and text embeddings is used to identify the most likely target object.

This modular design allows our method to operate in a zero-shot setting without fine-tuning, while maintaining competitive accuracy on standard benchmarks. More importantly, the lightweight nature of the framework makes it highly efficient and deployable for real-time robotic tasks where both speed and precision are critical.

II. RELATED WORKS

Visual grounding has become a critical task in vision-language research, particularly in robotics, where it enables agents to localize objects in complex environments based on natural language instructions. Traditional supervised methods often rely on region proposal networks and cross-modal fusion, but they suffer from limited generalization and high computational overhead. The advent of CLIP [1] introduced a new paradigm of zero-shot grounding by leveraging contrastive pretraining on large-scale image-text pairs. However, CLIP’s global feature alignment often struggles with precise object localization. To address this, recent approaches have explored visual prompting strategies to guide CLIP’s attention. Methods like CPT and ReCLIP utilize techniques such as region cropping, colored boxes, and Gaussian blur to highlight candidate regions [2]. RedCircle introduces circular prompts to simulate foveal vision [3], while FGVP (Fine-Grained Visual Prompting) demonstrates that applying blur outside a segmentation mask can significantly enhance semantic focus and grounding accuracy. Complementary to these, MaskCLIP [4] and MaskCLIP++ [5] improve regional alignment by combining masked self-distillation and consistency constraints. Furthermore, recent studies have emphasized the importance of adaptive prompting and label mapping: ILM-VP [6] co-optimizes label mappings with prompt design to enhance task transferability, while DAM-VP [7] uses a clustering-based meta-prompt framework to handle high-diversity datasets more effectively. In addition, applications of visual prompting in specialized domains such as medical imaging have shown its broader utility; for example, Denner et al. [8] demonstrate that adding visual markers to radiological scans helps CLIP focus on pathological regions, improving interpretability and zero-shot performance. Our work builds on these insights by proposing a lightweight, two-stage framework that integrates semantic-aware candidate extraction with visual prompting techniques to achieve accurate, efficient, and real-time object

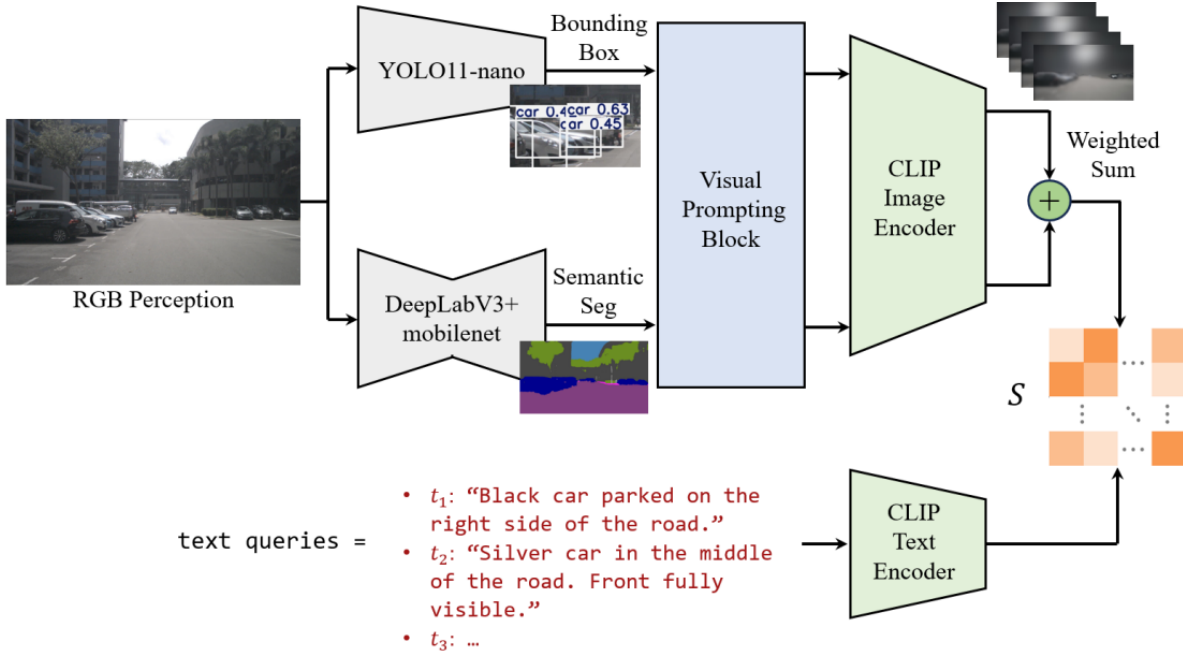


Fig. 1: General architecture of the proposed two-stage semantic-aware visual prompting framework. The system uses YOLO11-nano and DeepLabV3+ to extract bounding boxes and semantic segments respectively, and applies visual prompting before cross-modal alignment via CLIP.

grounding in robotic systems without requiring task-specific training or fine-tuning.

III. METHODOLOGY

A. Framework Overview

Our goal is to ground natural language queries in first-person RGB images by identifying the referred object in the scene. We design a lightweight and modular two-stage framework to tackle this problem, as shown in Fig. 1. The first stage extracts candidate object regions based on semantic and structural cues, while the second stage enhances each region via visual prompting for better alignment with the input language query.

This modular design allows us to combine structured visual perception with the powerful representation ability of CLIP without any task-specific fine-tuning. Compared to monolithic models, our pipeline is interpretable, efficient, and generalizable to various grounding scenarios.

B. Stage 1: Semantic-Aware Candidate Extraction

To reduce the complexity of full-image grounding, we first extract a set of potential object candidates that may match the input expression. We use two complementary perception modules:

- **YOLO11-nano**: provides bounding boxes for salient object-like regions in the scene. Its speed and compact size make it suitable for robotics.
- **DeepLabV3+ with MobileNet**: generates semantic segmentation maps to provide scene-level understanding and guide object disambiguation.

To fuse these two outputs, we filter object proposals based on semantic consistency: only bounding boxes that overlap with meaningful class regions (e.g., car, cup, person) are retained. This fusion strategy helps filter out noisy or redundant boxes (e.g., background clutter), yielding high-quality candidate regions $\{x_i\}_{i=1}^N$.

Each candidate x_i is passed to the next stage for targeted enhancement via visual prompting.

C. Visual Prompting Module

The key innovation of our method lies in the design of a dual-path visual prompting block, which generates two transformed inputs for each candidate region:

- **Region-Cropped View (x_i^c)**: A tightly-cropped patch centered on the candidate bounding box. This view captures fine-grained local features such as color, texture, or shape.
- **Blurred Contextual View (x_i^b)**: The full image is modified with a spatially-varying Gaussian blur:

$$x_i^b(p) = \begin{cases} x(p), & \text{if } p \in \text{ROI}_i \\ \mathcal{G}_\sigma(x(p)), & \text{otherwise} \end{cases}$$

where $\mathcal{G}_\sigma$ denotes Gaussian blur with increasing intensity for farther or unrelated regions.

This design draws inspiration from human foveal vision—humans focus sharply on objects of interest while peripheral vision remains blurred but semantically useful. By

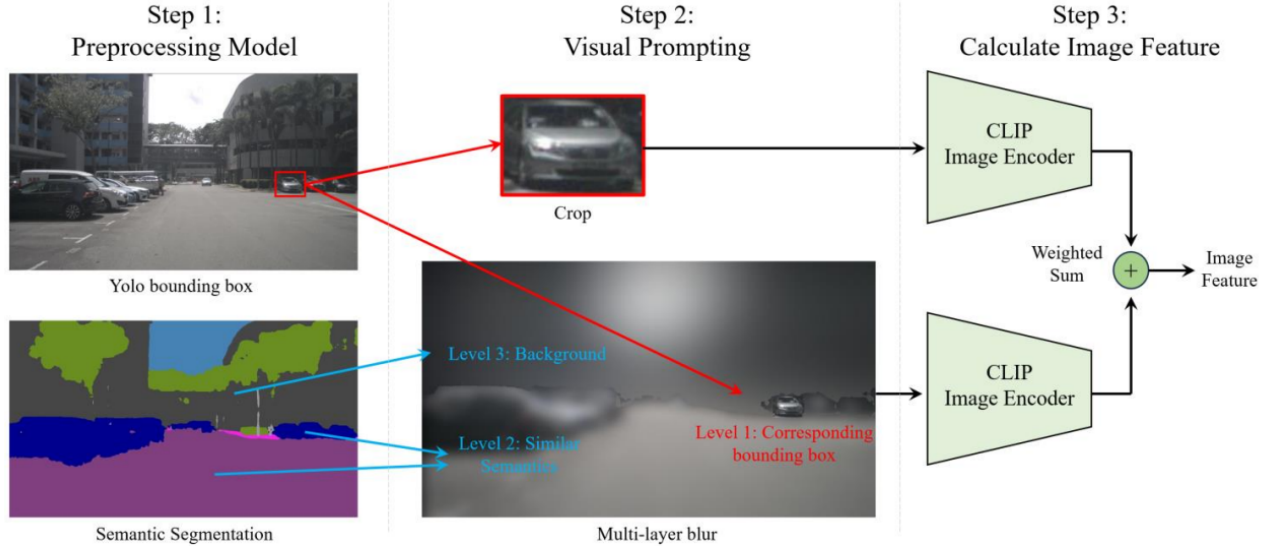


Fig. 2: Detailed illustration of the visual prompting process. Step 1: Candidate extraction using bounding boxes and semantic segmentation. Step 2: Application of cropping and multi-level blur to enhance focus. Step 3: Image features are computed through CLIP and compared with text features.

combining x_i^c and x_i^b , we allow the model to capture both local appearance and global scene context.

These two views are encoded by CLIP’s image encoder, and their embeddings are fused as:

$$f_i = \lambda \cdot f_{\text{clip}}(x_i^c) + (1 - \lambda) \cdot f_{\text{clip}}(x_i^b)$$

where $\lambda \in [0, 1]$ balances attention between fine detail and context (empirically set to 0.5 in our experiments).

D. Text Encoding and Cross-Modal Matching

Given a natural language expression t , we obtain its embedding f_t using CLIP’s pretrained text encoder. The similarity score between f_t and each candidate f_i is computed using cosine similarity:

$$S_i = \frac{f_i \cdot f_t}{\|f_i\| \cdot \|f_t\|}$$

The final grounding prediction corresponds to the candidate region with the highest similarity score:

$$x^* = \arg \max_i S_i$$

This simple yet effective matching mechanism enables our system to operate in a fully zero-shot setting. Since CLIP is pretrained on diverse web-scale data, it can generalize to open-vocabulary object descriptions without requiring explicit class labels or task-specific adaptation.

IV. EXPERIMENTS

We conduct comprehensive experiments to evaluate the effectiveness of our proposed two-stage semantic-aware visual prompting framework. The goal is to assess the model’s visual grounding accuracy across multiple datasets and compare its performance with state-of-the-art methods.

A. Datasets

We evaluate our model on three widely-used benchmarks for referring expression comprehension: RefCOCO, RefCOCO+, and RefCOCOg. These datasets consist of real-world images paired with natural language referring expressions describing specific target objects.

- **RefCOCO:** Contains short expressions, often involving spatial language such as “the man on the left”.
- **RefCOCO+:** Expressions focus on appearance and remove most spatial terms, increasing the reliance on visual details.
- **RefCOCOg:** Includes longer, more complex expressions collected in non-interactive settings.

B. Evaluation Protocol

Following standard practice, we use **accuracy** as the primary evaluation metric, defined as the percentage of samples where the predicted bounding box has an Intersection over Union (IoU) greater than 0.5 with the ground truth. This threshold-based correctness reflects whether the model successfully localized the referred object.

C. Baseline Methods

We compare our method with several representative baselines:

- **CPT-adapted:** A contrastive pretraining-based method using colorful boxes and basic prompts.
- **ReCLIP:** Employs region cropping and blur-based visual prompting for image-text alignment.
- **RedCircle:** Introduces circular prompts to highlight object regions within a global view.
- **FGVP:** Fine-Grained Visual Prompting uses precise instance-level masks generated by the Segment Anything Model (SAM) to blur non-target regions.

TABLE I: Accuracy (%) on RefCOCO, RefCOCO+, and RefCOCOg. A prediction is correct if IoU with ground truth exceeds 0.5. Visual Prompt: P = Crop, B2 = Colorful Box, B4 = Blur Reverse, C1 = Circle, D4 = Blur Reverse Mask.

Method	Visual Prompt	RefCOCO			RefCOCO+			RefCOCOg	
		val	testA	testB	val	testA	testB	val	test
CPT-adapted	B2	23.2	21.4	27.0	23.9	21.6	25.9	22.3	23.7
CPT-adapted	P B2	40.1	40.0	44.0	40.1	41.8	41.1	51.3	51.9
ReCLIP	P B4	45.8	46.0	47.1	40.0	50.1	45.1	59.3	59.0
RedCircle	P C1	43.9	46.2	44.3	47.9	43.1	47.3	57.3	56.3
Hierarchical Semantic Model	P D4	46.2	48.2	45.7	48.3	51.9	49.4	59.0	58.6
FGVP	P D4	52.0	55.9	48.8	50.4	60.4	46.7	62.1	61.9

Note: The results of baseline methods are quoted from the *Fine-grained Visual Prompting* paper [2].

D. Quantitative Results

Table I presents accuracy scores across all datasets. Our method achieves highly competitive results while maintaining significantly lower computational demands compared to FGVP.

E. Analysis

The quantitative results demonstrate that our proposed method achieves highly competitive performance across all three datasets, while maintaining a significantly lower computational burden compared to methods like FGVP.

Although FGVP achieves the highest accuracy in most settings, its performance gain comes at the cost of heavy computational overhead. This is primarily due to its reliance on the SAM for instance-level mask generation. SAM introduces high GPU memory consumption and long inference times, making FGVP less suitable for real-time applications, especially in robotic systems with limited hardware resources.

In contrast, our approach avoids the use of external segmentation models by leveraging lightweight semantic segmentation (DeepLabV3+) and object detection (YOLO11-nano) to generate candidate regions. This modular pipeline enables efficient preprocessing while preserving semantic richness, allowing the model to focus only on relevant object candidates.

Furthermore, our **visual prompting block** plays a crucial role in improving grounding performance. The combination of *region cropping* and *multi-level Gaussian blur* enhances the contrast between the target object and its surrounding context. The cropped version preserves fine-grained local features such as shape and color, while the blurred version encodes spatial layout and scene-level cues. By fusing both views, we provide a semantically-aware and spatially-guided visual input to the CLIP image encoder, leading to better alignment with textual descriptions.

Another key factor behind the effectiveness of our method lies in its compatibility with CLIP. CLIP is trained on a large-scale web dataset with diverse language-image pairs, but it is not explicitly designed for grounding tasks. Our prompting strategy adapts CLIP’s capability to this task without any fine-tuning, effectively simulating the “focus” mechanism that

human attention performs when grounding objects in a scene. This enables strong zero-shot generalization, as demonstrated by the consistent accuracy across datasets with varying linguistic styles and visual complexity.

Lastly, the simplicity and flexibility of our method make it easy to deploy and extend. It can be integrated into real-world robotic systems with minimal cost, and can also benefit from future improvements in base encoders (e.g., newer CLIP variants or segmentation models) without altering the core architecture.

In summary, our model strikes a favorable balance between accuracy, efficiency, and deployment practicality. It demonstrates that with careful design of visual prompts and candidate selection, high-quality object grounding can be achieved without expensive computation or supervised training.

V. APPLICATION

To validate the practical relevance of our method, we demonstrate its application in two real-world robotics scenarios: autonomous driving and instruction-guided manipulation.

A. Autonomous Driving

In autonomous driving scenarios, our framework enables zero-shot localization of vehicles by interpreting both visual appearance and spatial-temporal semantics embedded in natural language descriptions. For example, queries like “A *Jeep with a spare tire mounted on the back*” focus on distinctive shape and part-level details, while descriptions such as “A *car in the middle of the road. Front fully visible. The car is driving ahead*” or “A *car is parked on the left side of the road, facing left*” convey motion status, orientation, and positional context within the driving scene. The results are shown in Fig 3. Our two-stage semantic-aware visual prompting pipeline effectively grounds such queries in real-time RGB streams from a first-person perspective, capturing both static attributes and action-space reasoning. This highlights the model’s ability to comprehend nuanced semantics without any task-specific fine-tuning, making it well-suited for perception in dynamic urban environments.

B. Instruction-Guided Manipulator

We further demonstrate our method on a robotic arm tasked with object selection based on verbal instructions. Queries such as “carrot in the bowl”, “perrier soda”, and “coca cola” are used to locate the correct item from a cluttered tabletop scene. Results are shown in Fig 4. The visual prompting block enhances the region of interest and enables accurate object selection through CLIP-based similarity scoring.

These examples confirm that our lightweight and modular framework generalizes effectively to both navigation and manipulation tasks, without requiring task-specific fine-tuning, making it ideal for real-world deployment in robotics.

VI. CONCLUSION

In this paper, we introduced a lightweight, modular framework for object grounding in robotics based on two-stage semantic-aware visual prompting. By combining efficient candidate region extraction with dual-path visual prompting and CLIP-based cross-modal matching, our method achieves competitive accuracy across standard benchmarks (RefCOCO, RefCOCO+, RefCOCOg) without requiring any task-specific fine-tuning. The use of region cropping and spatially-aware Gaussian blur allows the model to simulate human-like attention, capturing both fine-grained visual details and contextual cues.

Our approach demonstrates strong generalization and zero-shot capabilities, making it particularly well-suited for resource-constrained robotic platforms that demand real-time performance. Through extensive experiments and real-world applications—including autonomous driving and instruction-guided manipulation—we showed that the framework enables accurate and robust object localization from natural language queries.

Future work may explore adaptive weighting strategies for view fusion, integration with temporal models for video grounding, and further optimization of the candidate extraction pipeline to enhance grounding precision under complex visual conditions.

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APPENDIX A QUALITATIVE RESULTS

To complement the quantitative analysis, we provide examples of our model’s grounding performance. These visualizations demonstrate the ability of our two-stage semantic-aware visual prompting framework to accurately localize referred objects based on diverse natural language expressions.



text_query= A car in the middle of the road; Front fully visible. The car is driving ahead.

text_query= A car is parked on the left side of the road, facing left.

(a) Grounding results for action-related semantics.



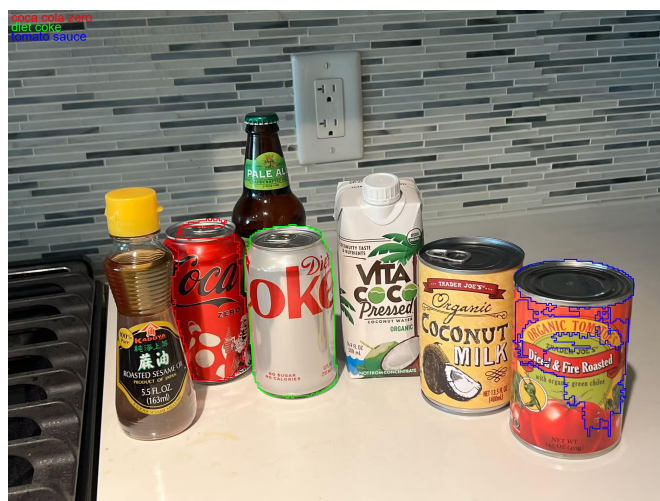
text_query= A Jeep with a spare tire mounted on the back.

(b) Grounding results for shape-related semantics.

Fig. 3: Application in autonomous driving. Example outputs showing the model grounding cars described by free-form queries.



(a)



(b)



(c)



(d)

Fig. 4: Instruction-guided manipulation using natural language queries. The system accurately grounds food or beverage items among distractors.